



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No.: MF F218, Unit: 3-1-4

Course Title: Transport phenomena in Manufacturing

Instructor-in-Charge: Dr. Aneesh A. M., Email: aneesh.am@pilani.bits-pilani.ac.in

Other Instructor: Mr. Sourabh Jain, Email: p20220438@pilani.bits-pilani.ac.in

- 1. Course description:** The course contents include: Introduction to Transport phenomena and its importance in manufacturing engineering, Fundamental concepts in Fluid statics and in motion, Heat and mass transfer theories, Conservation laws for Transport phenomena and solution methodologies, Important solutions and correlations in Transport phenomena, Radiation heat transfer and Special topics relevant to manufacturing such as Heat transfer with phase change, Simultaneous heat and mass transfer etc.

2. Scope and Objective:

This is a first level course aimed for imparting the fundamentals of fluid mechanics and heat transfer to students for solving transport phenomena problems pertaining to manufacturing processes.

3. Course learning outcomes:

CLO1. Introduce students to the fundamentals of fluid mechanics as well as heat and mass transfer required for the analysis of transport phenomena in manufacturing processes.

CLO2. Enable and motivate students to formulate and solve basic transport phenomena problems in manufacturing processes.

4. Text books and references:

TB1. Welty, Wicks, Wilson, Rorrer, Fundamentals of Momentum, Heat and Mass Transfer, 5th Edition, Wiley, 2010.

RB1. Yunus A. Çengel, John M. Cimbala, and Robert H. Turner, Fundamentals of Thermal Fluid Sciences, 4Ed, McGraw Hill Education, 2017.

RB2. Byron R. Bird, Warren E. Stewart, Edwin N. Lightfoot, Transport Phenomena, 2nd Edition, Wiley India, 2006.

RB3. William M. Deen, Analysis of Transport phenomena, 2nd Edition, Oxford University Press, 2013

RB4. Incropera FP and Dewitt DP, Fundamentals of Heat and Mass Transfer, 5th Edition, John Wiley and Sons, 2010.

5. Course Format & Weekly Workflow

The course will have 40 lectures covering the following 8 modules. The specific topics and their mapping TB is also provided below.

Wk. No	Mod. No.	Description	Ref.	Learning outcome
1-2	1	Introduction to Transport phenomena: Review of basic concepts in Fluid flow, heat transfer and mass transfer	TB1, RB1	To understand the relevance and scope of Transport phenomena.
3	2	Fluid statics: Concepts and applications	TB1, RB1	To understand and apply the concepts from fluid statics
4	3	Fluid kinematics: Concepts and applications	TB1, RB1	To understand and apply the concepts from fluid kinematics
5-6	4	Reynolds Transport Theorem, Mass, Momentum and Energy conservation equations. Solution methodologies	TB1, RB1	To understand the governing equations of transport phenomena and the solution methodologies
7-8	5	Steady and transient heat conduction	TB1, RB1	To understand and apply the concepts from heat conduction
9	6	Convective heat transfer: External flow	TB1, RB1	To understand and apply the concepts from external forced convection
10	6	Convective heat transfer: Internal flow	TB1, RB1	To understand and apply the concepts from internal forced convection
11	6	Natural convection over a vertical flat plate	TB1, RB1	To understand and apply the concepts from natural convection

12	7	Radiation heat transfer	TB1, RB1	To understand and apply the concepts from radiative heat transfer
13	8	Heat transfer with phase change, Simultaneous heat and mass transfer	Class notes	To introduce advanced transport phenomena concepts and its applications in manufacturing engineering

5.1. Mapping of course contents to the TB1:

- Basics of Transport phenomena: Fluids and the continuum (Section 1.1), Properties at a point (Section 1.2), Compressibility (Section 1.5), and Surface tension (Section 1.6). Newton's viscosity relation (Section 7.1), Non-Newtonian fluids (Section 7.2), and Viscosity (Section 7.3). Heat conduction (Section 15.1) and Thermal conductivity (Section 15.2). Molecular mass transfer (Section 24.1) and The diffusion coefficient (Section 24.2).
- Fluid statics: (All sections of Chapter 2)
- Fluid kinematics (All sections of Chapter 3 and Section 10.1),
- Conservation equations for Transport phenomena: Reynolds transport theorem (Class slides), Conservation of mass equation (all sections of Chap. 4), Conservation of momentum equation (all the sections of Chap. 5, 8, 9)
- Conservation of Energy equation (all sections of Chap. 6, 13 and 16),
- Fundamentals of heat transfer (all sections of Chap. 15)
- Conduction heat transfer: (17.1 to 17.3, 18.1)
- Convection heat transfer (19.1, 19.2, 19.6, 20.1, 20.2 and 20.3)
- Radiation heat transfer (23.1 to 23.6).

6. Evaluation Scheme:

Component	Weightage	Duration	Date and Time	Comments
Class discussions (short quizzes)	5%	TBA	Surprise type	Refer to point 8 for more details
Quiz 1 and 2	10%	40 min	TBA	Open*
Lab	25%		Monday, 11 AM to 1 PM.	Lab evaluation scheme will be announced during the first lab session.
Mid semester	25%	90 min	07/10, FN1	Open
Comprehensive	35%	180 min	08/12, FN	Open + Closed

*Note: During the open book examination, only those study resources approved by the IC will be permitted.

7. **Criterion for NC:** 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.
8. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components. The marks for class discussion will be awarded based on student's responses to random short quizzes shared during the middle of class.
9. **Chamber consultation hours:** Chamber: 6020-M, Monday 5:00 PM to 6:00 PM.
10. **Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for Mid-semester or comprehensive exam only.

Dr. Aneesh A. M.

Instructor-in-charge

MF F218